

MATHEMATICAL PHYSICS

2016–2017

MP204

Electricity and Magnetism

Time allowed: $1\frac{1}{2}$ hours

Answer **ALL** questions

This is a **SAMPLE** exam, roughly reflecting the general structure of the finals for 2016 – 2017.

Remember that **ALL** questions are to be answered.

If any figures are relevant, please include them with your solution! Drawing appropriate pictures often helps you arrive at the correct solution.

1. (a) The current density in some region is described by

$$\mathbf{J} = \gamma \left(\frac{4y}{x^2 + y^2} \hat{i} - \frac{4x}{x^2 + y^2} \hat{j} \right)$$

How is the charge density ρ in this region changing? If you use an equation related to the conservation of charge, name this equation explicitly.

[10 marks]

- (b) An infinitely long charged cylinder has radius R and uniform charge density ρ . Use Gauss's dielectric flux theorem to calculate the electric field inside the cylinder, at a distance $r < R$ from its axis. If you use any assumptions based on symmetry, state them clearly.

[20 marks]

- (c) An infinite wire carrying current I_1 runs along the y axis; the current flows from $y = -\infty$ to $y = +\infty$ through the origin. A square loop lies in the xy plane, with the four corners having coordinates (x_0, y_0) , $(x_0 + L, y_0)$, $(x_0 + L, y_0 + L)$, and $(x_0, y_0 + L)$. Current I_2 flows counterclockwise through the square loop.

Calculate the total force acting on the square loop due to the current in the long wire.

Reminders: (1) An infinitely long straight wire carrying current I_1 produces a magnetic field of strength $\mu_0 I_1 / (2\pi d)$ at a point at distance d from the wire. (2) A magnetic field B perpendicular to a wire of length L carrying current I_2 exerts a force $I_2 L B$ on the wire.

[20 marks]

2. (a) An electromagnetic system is described by the fields

$$\mathbf{E} = K_0 x \sin(\omega t) \hat{j} \quad \mathbf{B} = \frac{K_0}{\omega} \cos(\omega t) \hat{k}$$

Calculate the displacement current density $\mathbf{J}_D$.

Use Maxwell's equations to calculate the current density $\mathbf{J}$.

[15 marks]

- (b) A thin rod of length L lies along the x axis with one end at the origin and the other end at $(L, 0, 0)$. The rod carries a uniformly distributed positive charge Q .

Calculate the electric field generated by the charged rod at the point $(0, y_0, 0)$ on the y axis.

You might need the integrals

$$\int \frac{du}{(u^2 + a^2)^{3/2}} = \frac{u}{a^2 \sqrt{u^2 + a^2}}; \quad \int \frac{u du}{(u^2 + a^2)^{3/2}} = -\frac{1}{\sqrt{u^2 + a^2}}.$$

[23 marks]

- (c) A circular loop lies on a plane perpendicular to a uniform magnetic field of strength B_0 . The circular loop decreases its size over time: its radius is given by

$$r(t) = r_0 \left(1 - \frac{t}{t_0} \right).$$

Find the magnitude of the EMF induced in the loop.

[12 marks]